MYP 1 Mathematics: Assessment 4 (Term 2)

Criteria B (Investigating Patterns) & C (Communicating)

Section 1: Algebra and Sequences

Task 1: The Square Pattern

Consider the sequence of shapes made from square tiles:

Shape 1: 1 tile

Shape 2: 4 tiles

Shape 3: 9 tiles

1. Complete the table for the first 5 shapes:

   Shape (n): 1, 2, 3, 4, 5

   Tiles (T): 1, 4, 9, 16, 25

2. Identify the rule for this sequence. Is it linear or non-linear?

   Quadratic - non linear

3. Test your rule: Use your rule to calculate the number of tiles in Shape 10.

   100 tiles

4. Verify: If a shape has 144 tiles, which shape number (n) is it?

   n = 12

   +1
   16
   3
   48

5. Evaluate the expression 3x^2 + 5 when x = 4. Show your working.

   $3 \times 4^2 + 5 = 53$

Section 2: Shapes and Construction

48 49 50
S1 S2 S3

Task 2: Investigating Diagonals

A diagonal connects two non-adjacent vertices of a polygon. $\frac{n(n-3)}{2}$

1. Draw a pentagon (5 sides) and a hexagon (6 sides).

2. From ONE vertex, draw all possible diagonals.
3. Record the number of triangles formed inside each shape:
   - Quadrilateral (4 sides): 2 triangles
   - Pentagon (5 sides): 3 triangles
   - Hexagon (6 sides): 4 triangles

4. Discovery: Write a general rule (nth term) for the number of triangles (t) formed in a polygon with (s) sides. $s - 2$

5. Apply: If the sum of angles in one triangle is 180 degrees, use your rule

   1080

Answer Key & Marking Guidance

Section 1 Answers:

1. Table: Shape 4 = 16 tiles, Shape 5 = 25 tiles.
2. Rule: $T = n^2$ (Square numbers). It is non-linear.
3. Test: Shape 10 = 10^2 = 100 tiles.
4. Verify: $n^2 = 144 \rightarrow n = 12$. It is Shape 12.
5. Substitution: $3(4^2) + 5 = 3(16) + 5 = 48 + 5 = 53$.

Section 2 Answers:

3. Pentagon = 3 triangles; Hexagon = 4 triangles.
4. Rule: $t = s - 2$.
5. Octagon: 8 sides. Triangles = 8 - 2 = 6.
   Angle Sum = 6 * 180 = 1080 degrees.